

Robotics in Healthcare

Surgical Robots and Rehabilitation Robotics

A Research Report on Robotics Applications

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1. Introduction

Robotics has moved from industrial assembly lines into the operating theatre and the rehabilitation clinic. Healthcare is a demanding setting for any machine: tolerances are measured in millimetres, errors can be irreversible, and the end user is a vulnerable patient rather than a product on a conveyor. Despite this, medical robots have grown steadily since the 1980s, driven by advances in sensing, actuation, imaging, and, more recently, artificial intelligence (AI) [1], [3].

This report examines two of the most mature areas of healthcare robotics: **surgical robots** and **rehabilitation robots**. It traces their development, describes how the leading systems work, reviews the clinical evidence for their benefit, and discusses the technical, economic, safety, and ethical challenges that still limit adoption. It closes with emerging directions, notably autonomy and AI-assisted control.

2. Robotics in Surgery

2.1 Historical Development

The first documented use of a robot in surgery was in 1985, when a modified PUMA 560 industrial arm was used to position a needle for CT-guided stereotactic brain biopsy [2]. The appeal was precision: a robot can hold a trajectory without tremor or fatigue. In the early 1990s, orthopaedic systems such as ROBODOC were developed to mill the femoral canal for hip replacement with higher accuracy than manual rasping [3]. Soon after, research funded in part by the US military and NASA explored telepresence surgery, in which a surgeon operates remotely on a wounded soldier [3], [4].

Two commercial master-slave platforms, AESOP/Zeus and the da Vinci system, dominated the late 1990s. The da Vinci received US Food and Drug Administration (FDA) clearance for general laparoscopic surgery in 2000 and subsequently became the market leader [3]. In 2001, a team in New York performed a laparoscopic gallbladder removal on a patient in Strasbourg, France, using a robot controlled over a transatlantic fibre-optic link, the first demonstration of true remote robotic surgery [17].

2.2 How Surgical Robots Work

Most commercial surgical robots are *telemanipulators*, not autonomous machines. The surgeon sits at a console, views a magnified three-dimensional image from an endoscope, and moves hand controllers. The system translates these movements into scaled, tremor-filtered motion of miniature instruments inserted through small incisions. Typical features include wristed instruments with seven degrees of freedom, which restore the dexterity that straight laparoscopic tools lack, motion scaling that converts large hand movements into small instrument movements, and ergonomic seating that reduces surgeon fatigue [4], [14].

A second class comprises *image-guided or shared-control* systems, common in orthopaedics and neurosurgery. These register the patient's preoperative CT or MRI scan to the operating field, then constrain the surgeon's tool within a pre-planned boundary (a "haptic fence") so that bone is removed only where intended [3]. A third class includes steerable catheters and capsule-type devices for endovascular and endoluminal procedures.

2.3 Clinical Benefits and Evidence

The principal advantages reported for robot-assisted minimally invasive surgery are improved visualisation, greater precision in confined spaces such as the pelvis, and reduced surgeon fatigue. Compared with open surgery, patients commonly experience smaller incisions, less blood loss, and shorter hospital stays, in common with other minimally invasive approaches [4], [14]. Robotic assistance has been most widely adopted for prostatectomy, hysterectomy, and selected colorectal and cardiac procedures.

The evidence for superiority over conventional laparoscopy is, however, more mixed. Many comparative studies show similar long-term clinical outcomes, with advantages concentrated in technically difficult operations. A frequently cited economic analysis concluded that robotic surgery adds substantial cost and has diffused partly because of marketing and patient demand rather than proven outcome gains in every procedure [11]. Rigorous randomised evidence remains limited, and the learning curve for new operators is significant.

2.4 Safety and Adverse Events

Robotic systems introduce new failure modes. A retrospective analysis of 14 years of FDA adverse-event reports found thousands of reports, including device malfunctions, injuries, and deaths, with causes ranging from broken instrument fragments and electrical arcing to system errors and operator-related issues [13]. Importantly, such reports are voluntary and lack case-volume denominators, so they cannot be used to compute true complication rates. They do show the need for robust mechanical design, instrument life-cycle tracking, and thorough surgical team training.

3. Robotics in Rehabilitation

3.1 Motivation

Stroke, spinal cord injury, and neurological disease leave millions of people with persistent motor impairment. Recovery depends strongly on the amount, intensity, and repetition of task-specific practice, which is labour-intensive for therapists to deliver. Robots can provide high-dose, repeatable, quantifiable training and record objective data on patient performance [7], [15].

3.2 Upper-Limb and Gait Rehabilitation Robots

Early upper-limb devices such as MIT-MANUS used an impedance-controlled planar arm that guides the patient's hand through reaching tasks, assisting only as much as needed [15]. Later systems, including exoskeleton-type arms, add multi-joint movement and virtual-reality games to sustain motivation. For gait, treadmill-based systems with body-weight support and robotic leg orthoses, as well as over-ground and wearable exoskeletons, automate stepping practice that otherwise requires several therapists to supervise [9], [10].

Wearable exoskeletons such as ReWalk have enabled people with thoracic-level complete spinal cord injury to stand and walk with the aid of crutches, with improvements in mobility being offset by high cost and the need for supervised use [10]. Exoskeletons are currently used as both therapeutic tools in clinics and, in some cases, as personal mobility aids.

3.3 Clinical Evidence

Evidence for rehabilitation robots is encouraging but nuanced. An early randomised trial reported that robot-assisted reaching practice improved motor function after stroke relative to conventional therapy [7]. In the larger multi-site VA ROBOT trial, robot-assisted therapy using MIT-MANUS

did not significantly improve the primary measure of motor function at 12 weeks compared with usual care or intensive conventional therapy, although some secondary outcomes improved and costs were higher [8]. A Cochrane systematic review found that stroke patients who received electromechanical-assisted gait training in addition to physiotherapy were more likely to achieve independent walking, particularly in the first three months after stroke and among those who could not yet walk [9].

The overall picture is that robots are a useful way to deliver high-intensity practice rather than a replacement for therapists. Their main advantages are dose, consistency, and measurement; their effects appear to depend on patient selection, timing, and the quality of integration into a broader therapy programme.

4. Comparison of Application Areas

Aspect	Surgical Robots	Rehabilitation Robots
Primary goal	Improve precision, dexterity, and visualisation during operations	Deliver intensive, repeatable therapy to restore movement
Typical control	Surgeon-operated telemanipulation; shared-control in orthopaedics	Assist-as-needed or resistive control guided by patient effort
Example systems	da Vinci, ROBODOC, Zeus	MIT-MANUS, treadmill gait trainers, ReWalk
Strongest evidence	Reduced invasiveness; benefit in complex pelvic and prostate surgery	Electromechanical gait training after stroke
Main limitations	High cost, no haptic feedback, learning curve	Mixed trial results, cost, need for supervision

Table 1. Comparison of surgical and rehabilitation robots.

Table 1 summarises the principal differences between the two areas discussed above.

5. Challenges and Limitations

Cost and access. Surgical platforms involve high capital costs, annual service contracts, and expensive single-use instruments, which concentrates their availability in well-resourced hospitals [11]. Rehabilitation exoskeletons face similar barriers. Without evidence of clear value, health systems struggle to justify the investment, and this widens inequity between high- and low-income settings.

Lack of haptic feedback. Most commercial surgical robots provide limited or no true force feedback, so surgeons rely on visual cues to estimate tissue tension. Developing reliable, miniature, sterilisable force sensors remains an active research problem [4].

Safety, regulation, and liability. As systems gain autonomy, questions arise about who is responsible when something goes wrong: the surgeon, the manufacturer, or the software developer. Researchers have proposed graded levels of autonomy for medical robots, analogous to those for self-driving cars, along with regulatory frameworks that scale oversight with the level of autonomy [5].

Training and workflow. Operators need dedicated training, and operating-room staff must adapt to bulky equipment and new workflows. Learning curves can initially lengthen operation times and affect outcomes [14].

Data and cybersecurity. Networked robots collect sensitive patient data and could in principle be attacked remotely, so secure communication and privacy-preserving design are essential, especially for telesurgery over long distances [17].

6. Emerging Trends

6.1 Autonomy and Artificial Intelligence

Machine learning is being applied to surgical video for phase recognition, instrument tracking, and skill assessment, and may eventually support decision-making in the operating room [16]. In 2022, the Smart Tissue Autonomous Robot (STAR) performed laparoscopic intestinal anastomosis on live pigs with minimal human intervention, and the resulting sutures were reported to be more consistent than those made by expert surgeons or by teleoperation in the same study [6]. Such results are promising, but they were obtained in animal models under controlled conditions, and clinical translation will require extensive validation.

6.2 Telesurgery and Infection Control

Advances in low-latency networks make remote surgery increasingly practical and could extend specialist care to underserved regions [17]. The COVID-19 pandemic also showed the value of robots in healthcare more broadly, including disinfection, remote patient monitoring, and delivery of supplies, which reduce staff exposure to infection [12].

6.3 Wearable and Soft Robotics

Soft actuators and lightweight exosuits promise more comfortable and affordable rehabilitation devices that can be used at home. Combined with brain-computer interfaces and sensors that detect movement intention, they could enable closed-loop therapy that adapts to the patient in real time [10], [12].

7. Conclusion

Surgical and rehabilitation robots have progressed from experimental prototypes to established clinical tools. In surgery, the main gains are precision, visualisation, and ergonomics, although evidence of superior outcomes is procedure-specific and costs are high. In rehabilitation, robots reliably deliver intensive, measurable therapy, with the strongest evidence in gait training after stroke. Key challenges remain in cost, haptics, safety regulation, and equitable access. The next decade will likely be shaped by AI-assisted and partly autonomous systems, telesurgery, and wearable devices, but these must be introduced alongside rigorous clinical trials and clear ethical and legal frameworks so that benefits reach patients safely and fairly.

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